Verify that if $h^2 c > 4$ the Solutions for w are imaginary in equation (2.14)

$$2 - 2\cos(wh) = h^2 c > 4$$

$$-2\cos(wh) > 2$$

$$-(e^{iwh} + e^{-iwh}) > 2$$

$$e^{iwh} + e^{-iwh} < -2$$

Only $w = -ia + b$ will work as $|e^{iwh} + e^{-iwh}|$ is always ~~smaller~~ 2 and phase -1 gives a minimum of -2 which is ~~smaller~~ not smaller than -2

$$e^{ah} e^{ibh} + e^{-ah} e^{-ibh} < -2$$

for correct phase $b = \pi$

$$e^{ah} + e^{-ah} > 2$$

$$2\cosh(ah) > 2$$

$$\cosh(ah) > 1$$

$$ah > 1$$

$$a > \frac{1}{h}$$

so

$$w > -i\frac{1}{h} + \pi$$

$$x(t) = e^{iwt} = e^{\frac{t}{h}} e^{i\pi t}$$

$$\lim_{t \to \infty} (e^{\frac{t}{h}}) = \infty$$ thus unstable